

The Single-Site T -Matrix of a Legendre Cell in Closed Form

Elastic Waves, Every Power of the Frequency on Six Constants

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Draft, 7 October 2026

Abstract

A volume-integral method for elastic waves divides a heterogeneous medium into cells, each with a single-site T -matrix that maps the field incident on it to the sources it radiates. The governing equation is written in displacement; how the strain inside a cell is represented is a modelling decision, and the original approximation takes it as uniform. We take the cell whose displacement and strain are independent Legendre expansions over the cube, tested against the same polynomials (a Galerkin cell), because of the representations in use it gives the most accurate coupling between cells: its interactions are exact double integrals over both cells, where an expansion about the cell's centre treats the receiving cell as a point. For the linear Legendre cell with a uniform contrast we give the single-site T -matrix in closed form at every power of the wavenumber. Its self-interaction is a power series in $k_S h$, h the half-width, whose coefficients are moments of powers of the distance over the cell. Every coefficient is exact: an even power of $k_S h$ is a combination with rational coefficients of six constants, 1 , π , $\sqrt{2}$, $\sqrt{3}$, $\ln(1 + \sqrt{2})$ and $\operatorname{arsinh}(1/\sqrt{2})$, and an odd power, which carries the radiation, is rational. The first of these is the radiation damping of a point force, recovered exactly. On the irreducible representations of the cube group the 36 unknowns of the cell separate into blocks of at most four, and the T -matrix becomes a handful of closed-form matrices per power of $k_S h$. Through $(k_S h)^6$ the closed form reproduces the T -matrix computed by quadrature to 3.5×10^{-9} at $k_S h = 0.2$, falling as $(k_S h)^7$ as the next term predicts, and each of its 223 moments to 1.7×10^{-15} .

1 Introduction

A volume-integral method for elastic scattering replaces a heterogeneous medium by cells, here cubes, and couples them through the Green's tensor of the background. Each cell is characterised by its single-site T -matrix: the map from the field incident on it to the sources it radiates when it is alone. The coupled system is then the multiple-scattering problem of the cells [1–4], and everything that a cell contributes to the solution, beyond its coupling to the others, is in its T -matrix.

How the field inside a cell is represented decides what that T -matrix is. The governing equation

is written in displacement, and the strain enters it only through the stiffness contrast; how the strain is represented within a cell is therefore a modelling decision. The original approximation, of the equivalent-inclusion method [5, 6] and of the coupled-dipole method for elastic waves [7], takes it as uniform. That approximation can be relaxed in two ways. In the first, the displacement is expanded about the cell's centre and the strain derived from it; the equation and its derivatives are imposed at the centre, which is natural for a single scatterer, whose T -matrix is then a property of the body alone [8]. In the second, the strain is an unknown in its own right, with its own polynomial expansion over the cell, and the equation is tested against the cell's polynomials (a Galerkin cell). A strain derived from each cell's projected displacement would jump across the faces between cells and leave a spurious source there; an independent strain does not.

This paper takes the second, because it gives the more accurate representation of the coupling between cells. Its interactions are exact double integrals over both cells, which are evaluated to round-off even for cells that touch [9]; an expansion about the centre treats the receiving cell as a point. On a cube cut into n^3 cells, where the field follows the corners, the first-moment Legendre cell of 36 unknowns is seven to twelve times more accurate than the third-gradient expansion about the centre with 60; on a smoothly graded sphere both converge at fourth order, and the Legendre cell is the more accurate for a given number of unknowns [8]. And a Galerkin cell holds the projection of the field and of the medium, which is what makes its error a projection error, known before the solve [10].

The single-site T -matrix of the linear Legendre cell was given in Nestor [9] in its static limit, entry by entry on five constants. Here it is given at every power of the frequency. The self-interaction of the cell is a power series in $k_s h$ whose coefficients are universal moments of powers of the distance over the cell [9]; we evaluate every one of them exactly (§3), find that each power of $k_s h$ is a combination with rational coefficients of the same six constants, and that each odd power, which carries the radiation, is rational (§4). The cube group reduces the 36 unknowns to blocks of at most four (§5), on which the T -matrix is a few closed-form matrices per power. Section 6 gives the checks, each against an evaluation that shares no step with the closed form.

2 The cell and its single-site matrix

The governing equation. With time dependence $e^{-i\omega t}$, split the density and the stiffness into the background and the contrast, $\rho = \rho_0 + \Delta\rho$ and $c = c^0 + \delta c$, $\delta c_{bcde} = \Delta\lambda \delta_{bc}\delta_{de} + \Delta\mu(\delta_{bd}\delta_{ce} + \delta_{be}\delta_{cd})$. The contrast acts on the background as a body force, $\omega^2 \Delta\rho u_b + \partial_c m_{bc}$ with $m_{bc} = \delta c_{bcde} \varepsilon_{de}$, and with $\mathbf{G}$ the background's Green's tensor the displacement is

$$u_a(\mathbf{x}) = u_a^0(\mathbf{x}) + \int G_{ab}(\mathbf{x} - \mathbf{y}) \omega^2 \Delta\rho u_b(\mathbf{y}) d^3 y + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x} - \mathbf{y}) m_{bc}(\mathbf{y}) d^3 y, \quad (1)$$

the moment radiating through the receiver derivative of $\mathbf{G}$ [8]. The strain enters only through m . Differentiating Eq. (1) gives the strain at $\mathbf{x}$, and the pair $\psi = (u, \varepsilon)$, nine components with the strain in engineering Voigt form, satisfies one Lippmann–Schwinger equation with a 9×9 kernel

and the contrast operator $\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C})$, $\Delta \mathcal{C}$ the Voigt form of δc [8].

The strain model. Table 1 sets out the three representations of the strain that this series meets. They differ in the unknowns, and the test of the equation follows from the unknowns: derivatives at a point are fixed by imposing the equation and its derivatives there, moments over a cell by testing against the cell's own polynomials. This paper is about the third.

Table 1. How the strain is represented within a cell, and the test that follows. The words *collocation* and *Galerkin* describe only the test.

strain model	unknowns of a cell	strain and displacement	test
uniform (the original approximation)	displacement and strain at the centre	strain uniform	at the centre
derived	$\partial^p u$ at the centre, $ P \leq \ell$	strain the gradient of a Taylor displacement, degree $\ell - 1$	the equation and its derivatives at the centre
independent (this paper)	Legendre moments of displacement and strain to degree p	independent, both degree p	against the cell's polynomials (Galerkin)

The linear Legendre cell. A cube $[-h, h]^3$ with local coordinates $\xi = (\mathbf{x} - \mathbf{x}_c)/h$ carries $\psi(\mathbf{x}) = \sum_b \psi_b L_b(\xi)$ over the four functions $L_b \in \{1, \xi_0, \xi_1, \xi_2\}$, each ψ_b a nine-component vector: 36 unknowns. With the contrast uniform in the cell the source density $\Delta \psi$ is a polynomial on the same four functions. Testing Eq. (1) and its derivative against each L_a gives the single-site equation [9]

$$(\mathbf{M} - \mathbf{K}\Delta)\psi = \langle L, \psi^0 \rangle, \quad \mathbf{M}_{ab} = \langle L_a, L_b \rangle \mathbf{I}_9, \quad (2)$$

with $\mathbf{M} = \text{diag}(8h^3, \frac{8}{3}h^3, \frac{8}{3}h^3, \frac{8}{3}h^3) \otimes \mathbf{I}_9$ and $\mathbf{K}$ the self block, the kernel integrated against L_a over the cell as receiver and L_c over the cell as source. The single-site matrix maps the Legendre coefficients of the incident field to the source moments $\sigma_a = \langle L_a, \Delta \psi \rangle$, the mean force and stress and their first moments,

$$\mathbf{T}_{36} = \mathbf{M}\Delta(\mathbf{M} - \mathbf{K}\Delta)^{-1}\mathbf{M} = \mathbf{M}\Delta(\mathbf{I} - \mathbf{N}\Delta)^{-1}, \quad \mathbf{N} = \mathbf{M}^{-1}\mathbf{K}. \quad (3)$$

Everything specific to the cube is in $\mathbf{N}$; the contrast enters only through Δ .

3 The self block as a series in the wavenumber

The series. The Green's tensor is $\mathbf{G} = (4\pi\mu)^{-1}[\delta_{ij}g_S + k_S^{-2}\partial_i\partial_j(g_S - g_P)]$ with $g_k = e^{ikr}/r = \sum_t (ik)^t r^{t-1}/t!$, so it is a sum over the powers r^m , $m \geq -1$, of two families, $\delta_{ij}r^m$ and $\partial_i\partial_j r^m$ [9]. The propagator adds up to two derivatives. Each entry of the self block is therefore a combination, with 9×9 tables $\mathbf{TA}_t$ and $\mathbf{TB}_t$ fixed by the propagator's structure, of the universal moments of the self cell

$$U_{ac}(m, \iota) = \int W_{ac}(\mathbf{s}) \partial^\iota r^m \big|_{\mathbf{s}} d^3s, \quad h = 1, \quad (4)$$

W_{ac} the cross-correlation of L_a and L_c over the cell, $\mathbf{s}$ the separation of a receiving and a radiating point, and ι a multi-index of at most four derivatives. Collecting the powers of the wavenumber,

$$\mathbf{K} = \sum_{n \geq 0} (k_S h)^n \mathbf{K}_n, \quad 4\pi\mu \mathbf{K}_n = i^n \left[\frac{\mathbf{A}_n}{n!} - (1 - \gamma^{n+2}) \frac{\mathbf{B}_n}{(n+2)!} \right], \quad \gamma = \frac{k_P}{k_S} = \frac{\beta}{\alpha}, \quad (5)$$

$$\mathbf{A}_n = \sum_{\iota} \tau \mathbf{A}_{\iota} h^{5-|\iota|} U(n-1, \iota), \quad \mathbf{B}_n = \sum_{\iota} \tau \mathbf{B}_{\iota} h^{7-|\iota|} U(n+1, \iota). \quad (6)$$

The term $n = 0$ is the static (Kelvin) block. The series is that of e^{ikr} and converges for every k ; at $k_S h \leq 0.2$ seven terms leave less than 4×10^{-9} of the T -matrix (§6). An entry of row type r and column type c (0 for displacement or force, 1 for strain or moment) scales as h^{5-r-c} , so the coefficients need be computed once, at $h = 1$.

The moments, exactly. For odd m the kernel r^m is singular or not smooth at $\mathbf{s} = 0$, which for the self cell is the centre of the support. The integral of Eq. (4) is then reduced, exactly as for two cells that touch [9]: the derivatives that make the kernel a distribution are moved onto W , whose second derivative carries plane deltas at its kinks; W is a polynomial on each of the eight boxes $[0, \pm 2]^3$ that meet at the singular point; and each piece is a box or face integral of a monomial times a power of the distance, reduced from volume to faces to edges by recurrences whose base cases are elementary. Carried out in exact arithmetic, every such moment is a closed form. For even m the kernel is a polynomial, $\partial^{\iota} r^m = 0$ for $|\iota| > m$, and the box integrals are rational. Through $(k_S h)^6$ the self block needs 223 moments, $m = -1$ to 7; each is derived symbolically (scripts/derive_legendre_cell_dynamic.py).

4 Six constants, and a rational radiation

The moments of Eq. (4) for odd m are built from the master integrals of the self cell's boxes, whose sides are 2 and whose corners lie at 0 and ± 2 . Their recurrences only multiply powers of $\sqrt{a^2 + b^2 + c^2}$ with $a, b, c \in \{0, 2\}$, which gives $\sqrt{2}$ and $\sqrt{3}$, and end in three base cases: $\operatorname{arsinh}(c/\sqrt{a^2 + b^2})$, which gives $\operatorname{arsinh} 1 = \ln(1 + \sqrt{2})$ and $\operatorname{arsinh}(1/\sqrt{2})$, and $\arctan(bc/a\sqrt{a^2 + b^2 + c^2}) = \pi/6$. So for every odd m , and for every derivative, $U(m, \iota)$ is a combination with rational coefficients of

$$1, \quad \pi, \quad \sqrt{2}, \quad \sqrt{3}, \quad \ln(1 + \sqrt{2}), \quad \operatorname{arsinh} \frac{1}{\sqrt{2}}. \quad (7)$$

By Eq. (6), $\mathbf{K}_n$ draws on $m = n - 1$ and $n + 1$. An even power of $k_S h$ therefore uses odd m and is a combination of the six constants; an odd power uses even m , polynomial kernels, and is rational. Divided by the 4π of Eq. (5), an even power is a rational combination of $1/\pi$, 1 , $\sqrt{2}/\pi$, $\sqrt{3}/\pi$, $\ln(1 + \sqrt{2})/\pi$ and $\operatorname{arsinh}(1/\sqrt{2})/\pi$, and an odd power a rational multiple of $1/\pi$. The derivation confirms both through the highest power computed: the constants that occur in $\mathbf{K}_0$, $\mathbf{K}_2$, $\mathbf{K}_4$ and $\mathbf{K}_6$ are exactly the six, and every entry of $\mathbf{K}_1$, $\mathbf{K}_3$ and $\mathbf{K}_5$ is rational. In the static strain blocks the two logarithms occur only in the combination $\Lambda = [\ln(1 + \sqrt{2}) - \operatorname{arsinh}(1/\sqrt{2})]/\pi$ of Nestor [9]; in the displacement blocks and at the dynamic orders they occur separately.

The radiation. The odd powers are the radiation of the cell. The first, $n = 1$, has a single non-zero entry: the uniform displacement against the mean force, the cell's monopole. In the reduced form of §5 it is

$$N_{T_{1u},1}[u, u] = \frac{i}{\mu} \left[\frac{2}{\pi} - (1 - \gamma^3) \frac{4}{3! \pi} \right] = \frac{2i(2 + \gamma^3)}{3\pi\mu}, \quad (8)$$

so that the term $k_S h^2 N_{T_{1u},1}$ of $\mathbf{M}^{-1} \mathbf{K}$ is $8h^3$ times $i k_S (2 + \gamma^3)/(12\pi\mu)$, the imaginary part of $\mathbf{G}(0)$: the radiation damping of a point force carrying the cell's total force, as it must be at lowest order. The next odd power, $n = 3$, is again rational; it enters A_{1g} , E_g , T_{1g} , T_{2g} and T_{1u} , and vanishes on T_{2u} , A_{2u} and E_u ; the fifth enters every representation.

5 The cube group

With the contrast uniform in the cell, $\mathbf{M} - \mathbf{K}\Delta$ commutes with the 48 signed permutations of the cube, acting on the 36 unknowns as the product of their action on the four functions and on the nine components, and it reduces on the irreducible representations of O_h [9]. The displacement's first moments $\xi_i u_j$ (9) span $A_{1g} + E_g + T_{1g} + T_{2g}$; the uniform strain (6) spans $A_{1g} + E_g + T_{2g}$; the uniform displacement (3) spans T_{1u} ; and the strain's first moments (18) span $3T_{1u} + 2T_{2u} + A_{2u} + E_u$. Altogether 15 even unknowns, $2A_{1g} + 2E_g + T_{1g} + 2T_{2g}$, and 21 odd, $4T_{1u} + 2T_{2u} + A_{2u} + E_u$. On a representation Γ of multiplicity n_Γ the operator is one $n_\Gamma \times n_\Gamma$ block repeated for each row of Γ ; the largest is T_{1u} , 4×4 . Table 2 gives one row of each, as integer vectors. The partners are chosen so that each set transforms as the same row of Γ ; the reduction is then exact, $\mathbf{K}V_\Gamma = V_\Gamma \mathbf{K}_\Gamma$ with no residual at any power (§6).

Table 2. One row of each irreducible representation, as combinations of the unknowns $\xi_i u_j$ ($L_b = \xi_i$ times the displacement component j), u_j , ε_{ij} and $\xi_i \varepsilon_{jk}$, axes 0, 1, 2.

Γ	vectors	M_Γ/h^3
A_{1g}	$\xi_0 u_0 + \xi_1 u_1 + \xi_2 u_2; \quad \varepsilon_{00} + \varepsilon_{11} + \varepsilon_{22}$	$\frac{8}{3}, 8$
E_g	$2\xi_0 u_0 - \xi_1 u_1 - \xi_2 u_2; \quad 2\varepsilon_{00} - \varepsilon_{11} - \varepsilon_{22}$	$\frac{8}{3}, 8$
T_{1g}	$\xi_1 u_2 - \xi_2 u_1$	$\frac{8}{3}$
T_{2g}	$\xi_1 u_2 + \xi_2 u_1; \quad 2\varepsilon_{12}$	$\frac{8}{3}, 8$
T_{1u}	$u_0; \quad \xi_0 \varepsilon_{00}; \quad \xi_0(\varepsilon_{11} + \varepsilon_{22}); \quad 2(\xi_1 \varepsilon_{01} + \xi_2 \varepsilon_{02})$	$8, \frac{8}{3}, \frac{8}{3}, \frac{8}{3}$
T_{2u}	$\xi_0(\varepsilon_{11} - \varepsilon_{22}); \quad 2(\xi_1 \varepsilon_{01} - \xi_2 \varepsilon_{02})$	$\frac{8}{3}, \frac{8}{3}$
A_{2u}	$2(\xi_0 \varepsilon_{12} + \xi_1 \varepsilon_{02} + \xi_2 \varepsilon_{01})$	$\frac{8}{3}$
E_u	$2(\xi_1 \varepsilon_{02} - \xi_2 \varepsilon_{01})$	$\frac{8}{3}$

On each representation

$$\mathbf{T}_\Gamma = M_\Gamma \Delta_\Gamma (\mathbf{I} - \mathbf{N}_\Gamma \Delta_\Gamma)^{-1}, \quad \mathbf{N}_\Gamma = \frac{1}{\mu} \sum_{n \geq 0} (k_S h)^n i^n \left[\frac{\mathbf{P}_{\Gamma,n}}{n!} - (1 - \gamma^{n+2}) \frac{\mathbf{Q}_{\Gamma,n}}{(n+2)!} \right], \quad (9)$$

with Δ_Γ the contrast on the vectors of Γ , which is $\omega^2 \Delta \rho$ on a displacement vector and a combination of $\Delta \lambda$ and $\Delta \mu$ on a strain vector, and $\mathbf{P}_{\Gamma,n}$, $\mathbf{Q}_{\Gamma,n}$ the reduced forms of $\mathbf{A}_n$ and $\mathbf{B}_n$ divided by $4\pi M_\Gamma$. Each of their entries is a rational vector on the six constants (§4); all of them,

through the highest power computed, are in `data/self_blocks_irreps.txt`. As an example, the uniform dilatation (A_{1g} , the strain vector) has $P_0 = -\frac{1}{3}$ and $Q_0 = -\frac{2}{3}$, so that its static entry is $\mu N = -\frac{1}{3} + \frac{1}{2}(1 - \gamma^2)\frac{2}{3} = -\gamma^2/3$, that is $N = -1/[3(\lambda + 2\mu)]$, the value for every shape [9]; all 36 static entries of the uniform and linear strain reproduce those of Nestor [9], coefficient by coefficient on its five constants. Its first dynamic correction is

$$\mu N_{A_{1g},2}[\varepsilon, \varepsilon] = -\frac{1}{2}\left[P_2 - \frac{1-\gamma^4}{12}Q_2\right], \quad \pi P_2 = \frac{4}{15} - \frac{4}{9}\pi + \frac{4}{15}\sqrt{2} - \frac{8}{15}\sqrt{3} + \frac{4}{3}\ln(1 + \sqrt{2}) + \frac{8}{3}\operatorname{arsinh}\frac{1}{\sqrt{2}}, \quad (10)$$

and likewise for Q_2 (the data file); $(k_S h)^2$ times this is the leading change with frequency of the cell's dilatational self-interaction.

6 Checks

Every step is checked against an evaluation that shares none of it (`scripts/derive_legendre_cell_dynamic.py`), with the background $\alpha = 5000 \text{ m s}^{-1}$, $\beta = 3000 \text{ m s}^{-1}$, $\rho = 2500 \text{ kg m}^{-3}$ and the contrast $\Delta\lambda = 2 \text{ GPa}$, $\Delta\mu = 1 \text{ GPa}$, $\Delta\rho = 100 \text{ kg m}^{-3}$. Table 3 gives the results.

Table 3. Checks of the closed form through $(k_S h)^6$. The reference for the series is the package's series of the self block (master integrals at forty digits for odd m , Gauss rules for even m), which itself agrees with quadrature of the singular integrals to 1.4×10^{-15} .

check	result
223 exact moments, $m = -1$ to 7, against the package's self block, exact series to $n = 6$, against the full series, $k_S h = 0.02, 0.05, 0.1, 0.2$	1.7×10^{-15} $1.1 \times 10^{-14}, 6.6 \times 10^{-12}, 8.4 \times 10^{-10}, 1.1 \times 10^{-7}$
apparent order of that difference	7.00
$\mathbf{T}_{36}$ from the exact series against $\mathbf{T}_{36}$ by quadrature, $k_S h = 0.05, 0.1, 0.2$	$2.1 \times 10^{-13}, 2.7 \times 10^{-11}, 3.5 \times 10^{-9}$
apparent order	7.00
the same truncated at $n = 4$	$5.9 \times 10^{-10}, 1.9 \times 10^{-8}, 6.1 \times 10^{-7}$; order 5.00
invariance of every representation's span under every $\mathbf{K}_n$	exact (no residual)
$\mathbf{T}_{36}V_\Gamma = V_\Gamma\mathbf{T}_\Gamma$, Eq. (9), worst over Γ , $k_S h = 0.05, 0.1, 0.2$	$4.4 \times 10^{-13}, 5.6 \times 10^{-11}, 7.2 \times 10^{-9}$
static uniform dilatation	$-1/[3(\lambda + 2\mu)]$ exactly
the 36 static strain entries against Nestor [9]	identical coefficients
radiation at $n = 1$	$\operatorname{Im} \mathbf{G}(0)$ exactly, Eq. (8)

The differences fall as $(k_S h)^7$, the first power omitted (and as $(k_S h)^5$ when the series stops at $n = 4$), so they are the truncation of the series and not an error of its terms. The closed form is that of the exact self block: what it omits at a given order is stated by the next term, which is itself in closed form.

7 Discussion

The single-site T -matrix of the linear Legendre cell is now known in closed form at every power of the frequency, on six constants, with its radiation rational and its first radiative term the point force's. Two things are not yet covered. The contrast here is uniform in the cell; a graded

contrast couples the even and odd unknowns and leaves only the operations that fix its gradient [9], so the reduction is weaker, though the moments are the same and the closed form carries over. And the field is linear; the quadratic cell of 90 unknowns needs the moments of monomials to degree four, which the same derivation provides. The comparison with the single site of the expansion about the centre, whose self-interaction is taken at the centre rather than averaged over the cell [8, 9], is to be condensed here from the earlier paper.

Data availability

The derivation and its checks are `scripts/derive_legendre_cell_dynamic.py`, which writes `data/self_moments_exact.pkl` (every moment, exact) and `data/self_blocks_irreps.txt` (every reduced entry, on the six constants), in the repository <https://github.com/tmnestor/MultipleScatteringCalculations>. The static blocks are `scripts/derive_graded_voxel_site_blocks.py`.

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